



Chapter I: Drag and Power Estimation of AUV

AUV requirements:

Operational depth: 300 m

Endurance: 6 Hrs

External Dia: 0.168 m

Payload: 20 kg

component	Make and Model No	Max POWER (VA)	Avg Power (VA)	Weight in Air (kg)	Weight in water (kg)	Size(LxWxH)	Component placement
NI cRIO controller	NI 9076	15	10	0.65	-	180 x 88 x 95	Inside enclosure
INS_SBG	SBG_ekonox	5	3	0.45		100 x 86 x 58	Inside enclosure
DVL	Navquest Micro	80	20	2.9	1.2	174 L x 126 D	
Scanning Sonar	Imagnex 881	10	7	1.5	0.6	182 L x 80 D	
Color Camera	ROS spectator	10	7	2.75	0.91	260 L x 155 D	
CTD- DO	RBR	5	5	1	0.45	490 L x 64 D	
LED light	Kongsberg	50	50	1.1	0.6	114 L x 95 D	
GPS	Parallox	3	2	0.1		35 x 35	Inside enclosure
Relays	Phenix	2	2	0.1		100 x 50 x 25	Inside enclosure
RF/WIFI transmitter	TBD	5	5	0.5		TBD	Inside enclosure
Underwater cables	assorted	0	0	2			
water Entry sensor	NIOT	2	2	0.05			Inside enclosure
wiring and other	assorted	5	5	0.5			Inside enclosure
HD camera with recorder	TBD	0		0.5		TBD	Inside enclosure
Battery				2.5			
Structures				2			
Total		192	118	18.6			

Length Sizing:

Pressure at the depth of 300 m,

$$P_{op} = \rho_{sw} * g * h_{op}$$

P_{op} – Hydrostatic pressure at operational depth

ρ_{sw} – Density of sea water

h_{op} – Operational depth

$$P_{op} = 1025 * 9.81 * 300$$

$$P_{op} = 3.01 \text{ MPa}$$

The AUV hull should with stand 3.01 MPa external pressure.

The available Aluminium pipe was chosen for hull material.

**Specifications:**

Dia	(*m)	0.168
Thickness	(*m)	0.007
Internal Dia	(*m)	0.154
Density	(Kg/m ³)	2800

The circumferential stress acting on the AUV hull at the depth of 300 m would be

$$\text{Circumferential Stress } \sigma = \frac{P_{op}d}{2t}$$

$$\sigma_{op} = \frac{3.01 * 10^6 * 0.168}{2 * 0.007}$$

$$\sigma_{op} = 36.2 \text{ MPa}$$

Yield stress of Al alloy

$$\sigma_{Al} = 214 \text{ MPa}$$

$$\sigma_{Al} > \sigma_{op}$$

This Al pipe can be used for hull since the circumferential stress at the operational depth of 300 m is lesser than the Aluminium pipe.

The Hull should accommodate the payload of 25 kg for the endurance of 6 hrs.

For neutrally buoyant condition,

$$\text{Weight of the Water Displaced} = \text{Weight of the hull} + \text{Pay load weight}$$

$$\text{Volume of Hull} * \text{Density of Sea water} = \text{Weight of the hull} + \text{Pay load weight}$$

$$\pi * R_{out}^2 * l * \rho_{sw} = \pi * (R_{out}^2 - R_{in}^2) * \rho_{pvc} * l + w_{payload}$$

$$l = \frac{w_{payload}}{\pi * [(R_{out}^2 (\rho_{sw} - \rho_{pvc}) + R_{in}^2 * \rho_{pvc})]} \text{ m}$$

Where,

l – length of the pipe required

$w_{payload}$ – pay load weight = 25 kg

R_{out} – Al pipe external radius = 0.084 m

R_{in} – Al pipe internal radius = 0.077 m

ρ_{sw} – Density of sea water = 1025 kg/m³

ρ_{Al} – Density of Al pipe = 2800 kg/m³

$$l = \frac{20}{\pi * [(0.084^2 (1025 - 2800) + 0.077^2 * 2800)]} \text{ m}$$



$$l = 1.56 \text{ m}$$

Drag or Resistance calculation:

Diameter of the AUV $D = 0.168 \text{ m}$

Length $L = 2.05 \text{ m}$

Dynamic Viscosity of water $\mu = 0.00108 \text{ Pa s}$

Density of sea water $\rho = 1025 \text{ kg/m}^3$

Velocity = 1.028 m/s or 2 knots

Fluid dynamic drag (stream lined body) comprises the following components

$$Re_l = \frac{\rho * v * L}{\mu}$$

$$Re_l = 2.00 \times 10^6$$

I. Drag based on the wetted surface area

$$\frac{C_{Dwet}}{C_f} = 1 + 1.5 * \left(\frac{d}{l}\right)^{1.5} + 7 * \left(\frac{d}{l}\right)^3$$

$$\frac{C_{Dwet}}{C_f} = 1.04$$

II. Drag based on the frontal area of the body

$$\frac{C_{Do}}{C_f} = 3 \left(\frac{l}{d}\right) + 4.5 * \left(\frac{d}{l}\right)^{0.5} + 21 * \left(\frac{d}{l}\right)^2$$

$$\frac{C_{Do}}{C_f} = 38.33$$

III. Skin friction drag(ITTC-1957)

$$C_f = \frac{0.075}{(\log_{10} Re_l - 2)^2}$$

$$C_f = 0.0040$$

The above skin friction factor should be multiplied by the form factor $(1+k)$.



$$(1 + k) = 1 + 1.5\left(\frac{d}{l}\right)^{3/2} + 7\left(\frac{d}{l}\right)^3$$

$$(1 + k) = 1.04$$

Total frictional drag co efficient

$$C_{fTotal} = C_f(1 + k)$$

$$C_{fTotal} = 0.0042$$

Pressure Drag:

$$\begin{aligned} \text{Pressure Drag Co efficient} &= \frac{C_{Dwet}}{C_f} * C_{fTotal} + \frac{C_{Do}}{C_f} * C_{fTotal} \\ &= 1.04 * 0.0042 + 38.33 * 0.0042 \end{aligned}$$

$$C_d = 0.164$$

Pressure Drag force can be calculated from the following relation

$$\begin{aligned} D_p &= \frac{1}{2} * \rho_{sw} * v^2 * S_{frontal} * C_d \\ D_p &= \frac{1}{2} * 1025 * 1.028^2 * \pi * 0.084^2 * 0.164 \\ \mathbf{D_p} &= \mathbf{1.975 \text{ N}} \end{aligned}$$

Skin friction Drag:

Skin friction drag force can be calculated by the following relation,

$$\begin{aligned} D_p &= \frac{1}{2} * \rho_{sw} * v^2 * S_{wet} * C_{fTotal} \\ S_{wet} &= \text{cylinder surface area (approximately)} \\ S_{wet} &= 2\pi rh + 2\pi r^2 \\ &= 2 * \pi * 0.084 * 1.56 + 2 * \pi * 0.084^2 \\ &= 0.87 \quad m^2 \\ D_s &= \frac{1}{2} * 1025 * 1.028^2 * 0.87 * 0.0042 \\ \mathbf{D_s} &= \mathbf{1.98 \text{ N}} \end{aligned}$$

Total Drag Force or Bare hull resistance

$$\begin{aligned} D &= D_p + D_s \\ D &= 1.975 + 1.98 \end{aligned}$$

$$D = 3.96N$$

Ref:

1. Fluid -Dynamic Drag theoretical, experimental and statistical information by Dr.S.F Hoerner
2. Design and construction of a Second Generation AUV by Michael R. Good
3. A Study on the Design Optimization of an AUV by Using Computational Fluid Dynamic Analysis by Taehwan Joung

Power Calculation (VT method):

Power required will be calculated from the bare hull resistance and appendage resistance (30% of Bare hull Estimated Power)

$$P_{bare\ hull} = D * V$$

$$P_{bare\ hull} = 3.96 * 1.028$$

$$P_{bare\ hull} = 4.07\ Watt$$

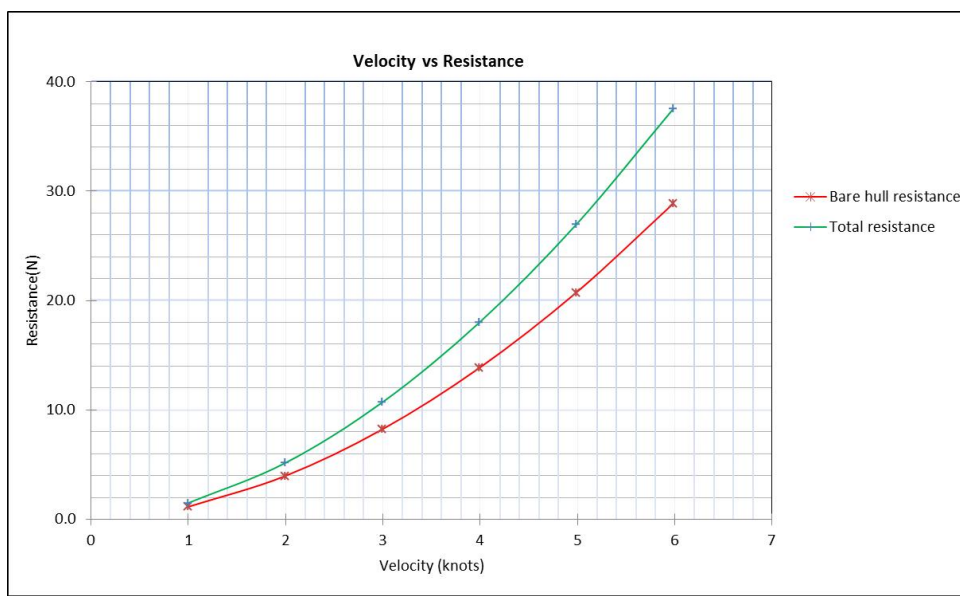
$$P_{appendage} = 0.3 * P_{bare\ hull}$$

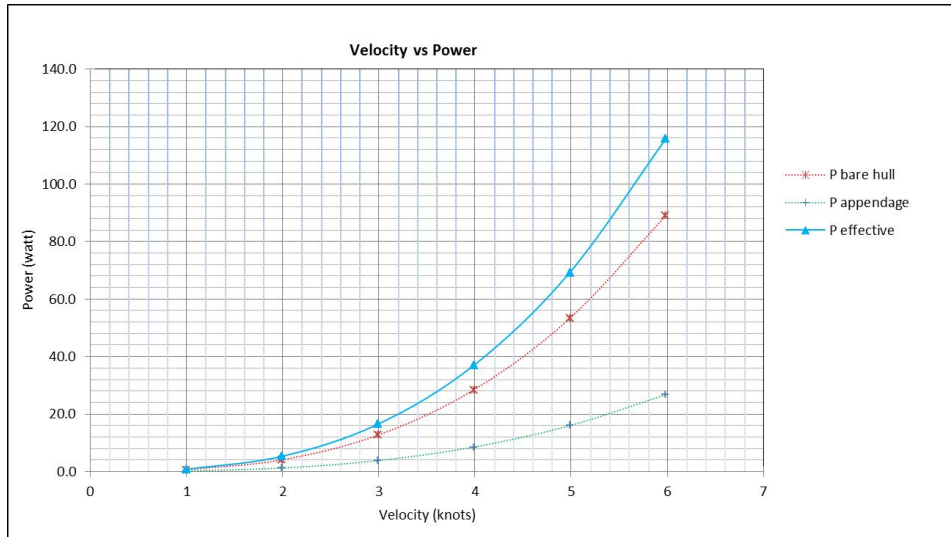
$$P_{appendage} = 1.22\ Watt$$

$$P_{Total} = P_{bare\ hull} + P_{appendage}$$

$$Effective\ Hull\ power\ P_E = 4.07 + 1.22$$

$$P_E = 5.29\ Watt$$





Ref:

1. Design Report-Guided Missile Submarine SSG(X) (2006), Aerospace & Ocean Engineering, Virginia Tech.

Excess Power required for Acceleration /Deceleration:

Vehicle mass: 36 kg

Acceleration /Deceleration Distance: 4 m (Approximately 2 times of vehicle length)

Speed: 2 knots (or) 1.03 m/s

Equation of Motion

$$v^2 = u^2 + 2as$$

$$\text{Initial velocity } u = 0$$

$$\text{Final velocity } v = 1.028 \text{ m/s}$$

$$\text{Distance } s = 4 \text{ m}$$

$$a = \frac{v^2}{2s}$$

$$a = 0.3 \text{ m/s}^2$$

Force Required

$$F_i = ma$$

$$F_i = 36 * 0.3$$

$$F_i = 10.8 \text{ N}$$

To accelerate/decelerate the AUV for a distance of 4 m, the minimum force required is 10.8 N or 1.10 kgf.

Total Force required F_t

= Inertial Force F_i + Barehull Resistance + Appendage Resistance N

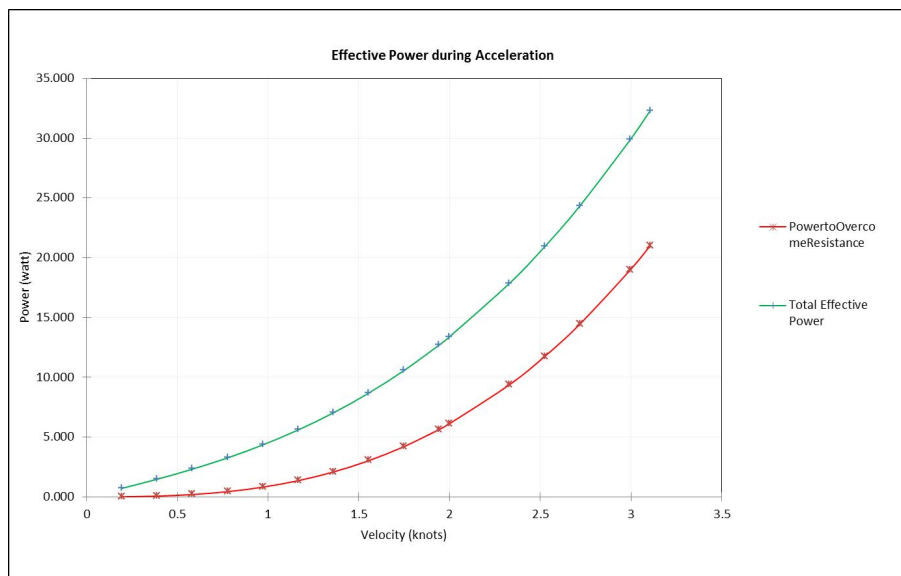
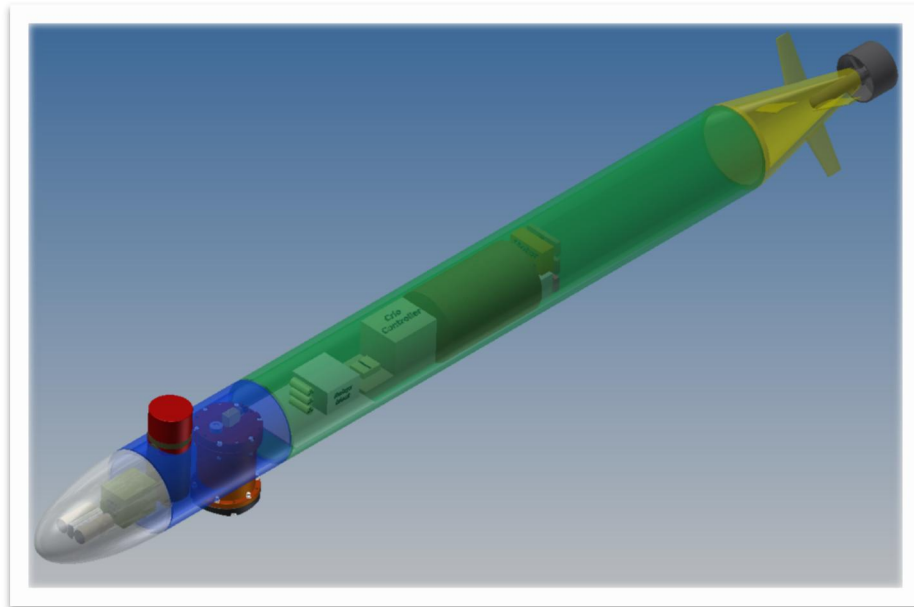
$$F_t = 10.8 + 3.96 + 3.96 * 0.3 \text{ N}$$

$$F_t = 15.948 \text{ N}$$

$$\text{Total Power required } P_t = F_t * v$$

$$P_t = 15.9 * 1.03$$

$$P_t = 16.42 \text{ Watt}$$





Length						
	Dia	0.168		TR	0.65	
	Length	2.05		AR	2	
	Width	0.168				
				Cmean	0.041	
	Area of Rudder	0.00402				
				Cr	0.050	
				Ct	0.033	
	Required Area (m2)	0.0068		b	0.083	
		0.0034				
				Re	4.04E+04	
	Jackson Method					
	Dia	0.17				
	Length	2.05				
	M	25.600				
	Required area (m2)	0.0135		TR	0.65	
				AR	2	
		0.0067		Cmean	0.058	
		Re	2.00E+06			
				Cr	0.070	
				Ct	0.046	
				b	0.116	